SS II
Chapter-6
THE CHANGING EARTH

1.Two types of Earth movements & their forces

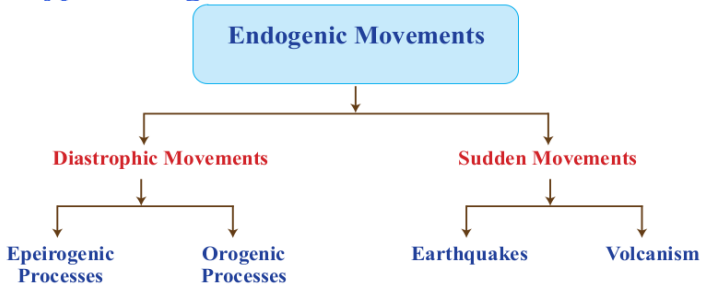
a. Endogenic Movements – Movements inside the Earth.

• **Forces:** Radioactivity, Earth's rotation, geothermal heat, convection currents.

b. Exogenic Movements –Movements on the Earth's surface.

• **Forces:** Slope, gravity, running water, groundwater, wind, glaciers, temperature changes, chemical actions.

2.Types Endogenic Movements



3.Diastrophic movements:

• Slow movements inside the Earth that lift or fold the crust.

- **Epeirogenic movements** –
  - Lift or lower large areas of land
  - Cause minimal deformation
- **Orogenic movements** – • Form mountains
  - Cause severe deformation

4. Volcanism-

• Opening in Earth through which magma, ash, gases come out.

5. Difference between magma and lava

• **Magma** – Hot, molten rock **i**nside the Earth.

• **Lava** – Magma that comes out to the Earth's surface.

6. Effects of volcanic activity

• Loss of life • Destruction of houses
• Stops economic activities • Makes people homeless

7. Only active volcano in India

• Barren Island (Andaman)

8. Precautions for volcano

• Avoid living near volcanoes * Control tourism
• Monitor volcanoes * Give warnings

9. Usefulness of volcanoes

• Lava soil is fertile * Geothermal energy
• Minerals * Tourism

10. How earthquakes occur when plates move?

• When two plates rub, friction increases.

• Pressure builds up and suddenly releases as strong seismic waves. This causes a tectonic earthquake.

11. Focus & Epicentre

• **Focus (Hypocentre):** The point inside the Earth where the earthquake starts.

• **Epicentre:** The point on the Earth's surface just above the focus and the first place to feel the quake.

12. Causes of earthquakes

• Plate movements, Volcanoes, Mining/explosions
• Big reservoirs

13. Problems in earthquake areas

• Loss of life • Buildings damaged
• Roads blocked • No water/food • Landslides, fires

14. Earthquake-Precautions (Before–During–After)

Before – Emergency kit, know safe places, avoid weak buildings

During – Stay calm, hide under table, go to open space

After – Give first aid, avoid damaged buildings, follow officials

15. Tsunami

• Huge waves caused by sea-floor earthquakes

16. Safety from tsunami

• If sea suddenly goes back → run to higher ground
• Hold something that floats

17. Exogenic Movements and Driving forces

• Weathering – temperature, gravity, chemicals
• Erosion etc. – water, wind, glacier
• Mass wasting – gravity

18.Types of Weathering

Physical weathering

• Rocks break due to heat, cold, or freezing water

Chemical weathering

• Water, oxygen etc. react with rocks

Biological weathering

• Plant roots, animals, decay of plants/animals

19. Human activities causing weathering

• Mining * Quarrying • Construction
* Deforestation • Farming *Pollution

20. Usefulness of weathering

• Forms soil * Separates minerals
• Enriches minerals *Makes mining easy

21.Erosion, Transportation, Deposition-Meanings

• **Erosion** – Wearing away of rocks
• **Transportation** – Moving eroded materials
• **Deposition** – Dropping materials in low places

22.Mass Wasting& types

• Movement of rocks/soil downhill due to gravity

Types: Landfall, rockfall, landslide, debris flow, land subsidence

23. Debris flow & Landslide

• **Debris flow** – Mud ,rocks , water move suddenly
• **Landslide** – Sliding of soil/rock down a slope

24. Urulpottal

• Landslide/debris flow in Kerala during heavy rain

25. Causes of Urulpottal

• Unscientific construction • Blocking natural streams
• Soil type • Land use • Heavy rain • Quarrying

26. Identifying landslide-prone areas

• Using Remote sensing & GIS
• Maps by GSI and NCESS available on KSDMA website

27. Safety during Urulpottal

During – Stay calm, hide under strong table/cot

After – Allow rescue vehicles, report fallen electric lines

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